

Can you please tell me the difference between websocket and API polling?

Sure! Here's a clear, concise comparison of **WebSockets vs. API Polling**:

1. How They Work

WebSocket

- Creates a **persistent, bidirectional connection** between client and server.
- After the handshake, both sides can push data anytime.
- Feels like a “live” connection—ideal for real-time apps.

API Polling

- Client repeatedly sends **HTTP requests at intervals** (e.g., every 5 seconds).
 - Server responds with the latest data each time.
 - Communication is **one-directional** (client → server).
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2. Performance & Efficiency

WebSocket

- **Highly efficient** for frequent updates.
- Uses one long-lived connection instead of many requests.
- Lower latency.

API Polling

- Can be inefficient—lots of repeated requests even if nothing changed.
 - Higher network and server load.
 - Latency depends on polling interval.
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3. Use Cases

WebSocket (best for real time)

- Chat apps
- Live dashboards
- Online gaming
- Stock/crypto tickers
- Collaborative editors

API Polling

- Situations where updates are **infrequent** or **not time-critical**.
 - Simple implementations.
 - When WebSocket support is lacking.
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4. Implementation Complexity

WebSocket

- More complex (connection handling, state, upgrades).
- Requires server support.

API Polling

- Very simple—just normal HTTP requests.
 - Works everywhere.
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5. Reliability & Scalability

WebSocket

- More stateful; needs proper scaling (sticky sessions, special load balancing).
- If done right, scales well for real-time use.

API Polling

- Stateless (easy to scale).
 - But heavy polling from many clients can overwhelm servers.
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Summary Table



Feature	WebSocket	API Polling
Connection	Persistent	Repeated requests
Direction	Bidirectional	Client → Server only
Latency	Very low	Depends on interval
Efficiency	High	Low for frequent updates
Complexity	Higher	Very simple
Ideal for	Real-time apps	Infrequent/low-priority updates

If you want, I can also explain:

- Long polling vs short polling vs WebSockets
- When to choose each
- Code examples (JS/Python)

